

CASE STUDY

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Microplastics distribution in water and sediment from river Uhl, Himachal Pradesh, India

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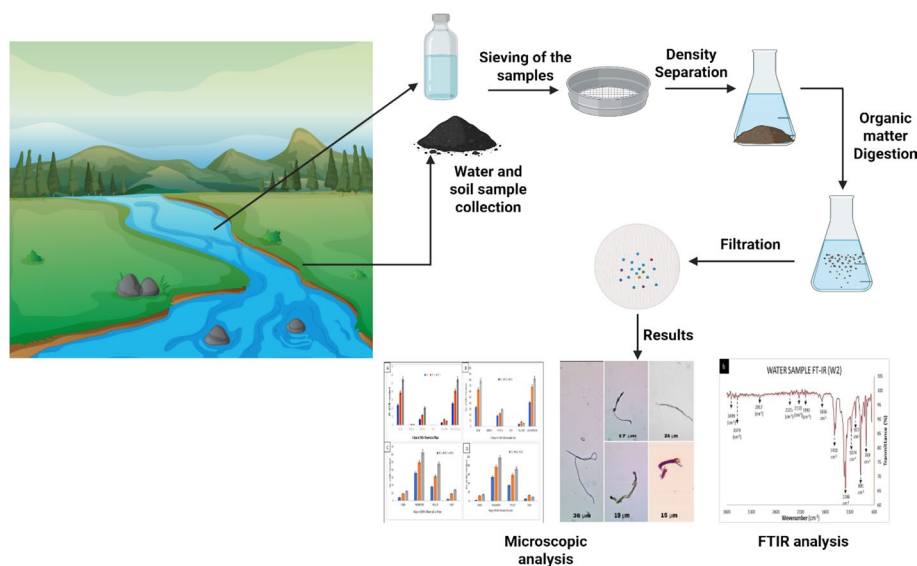
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Abstract

Microplastic (MP) pollution in freshwater ecosystems is a growing environmental concern, particularly in the fragile Himalayan River systems. This study investigated the occurrence, abundance, and characteristics of MPs in water and sediment samples collected from the River Uhl, originating from the Thamsar Glacier in the Dhauladhar range, Himachal Pradesh, India. Sampling was conducted at three sites representing varying degrees of anthropogenic influence and analysed through microscopic examination and Fourier-Transform Infrared (FT-IR) spectroscopy analysis. From each site, $n=3$ water samples (1 L each) and sediment samples (1 kg each) were collected. MP concentrations were expressed as particles L^{-1} for water and particles Kg^{-1} (dry weight) for sediment. MP abundance showed clear spatial variability, with lower concentrations at the relatively pristine site (50–100 particles L^{-1} or Kg^{-1}) and comparatively higher levels at downstream sites (100–500 particles L^{-1} or Kg^{-1}). Fragmented MP was the dominant morphotype in both water (83.16%) and sediment (87.61%), followed by fibres and films. Colour analysis revealed a predominance of colourless and blue particles in both matrices. FT-IR spectroscopy confirmed the presence of common polymer types and revealed site-specific spectral variations, indicating differences in polymer composition and degradation states influenced by local environmental conditions. The detection of MP even at sites with limited human activity underscores its widespread distribution and potential for transport. These findings underscore the need for continuous monitoring and targeted management strategies to mitigate MP pollution and protect freshwater ecosystems in the Himalayan region.

Keywords: Freshwater ecosystem, FT-IR characterization, Polymer characterization, Riverine pollution, Sediment, Spatial distribution



Graphical abstract**1 Introduction**

Plastics are long polymers of carbon and hydrogen, along with oxygen, nitrogen, chlorine, and sulphur. Some of the most commonly and frequently used polymers in our everyday life are Low-density polyethylene (LDPE), Polyvinyl chloride (PVC), High-density polyethylene (HDPE), Polypropylene (PP), and Polyethylene terephthalate (PET). LDPE is known for its flexibility and impact resistance, HDPE for its strength, PVC for its versatility, PET for its clarity, and PP for its heat resistance. These are used in the manufacturing of bottles, bags, clothing, packaging, paints, and PCPs [41]. Properties such as low manufacturing cost, water resistance, high strength-to-weight ratio, and high thermal and electric insulation properties mark its high production [13, 25].

Plastic had an annual production of nearly 445.25 million metric tons in 2025 and is expected to reach 516.18 million metric tons in 2035 [54]. Accordingly, over 380 million tonnes of plastic waste are produced globally, with India producing 9.4 million tonnes [42]. Consequently, India ranks 4th in production (9%) and consumes only 5% of total plastic [1, 42].

Such a huge production has aggravated the problem of plastic pollution. These have now infiltrated even the most isolated corners of the Earth, including the polar regions. In recent years, research into the presence of plastic waste in the Arctic and Antarctic oceans has significantly increased, leading to the unequivocal conclusion that it is a tangible, pressing, and significant problem [9, 29]. In the environment, the accumulated plastic may undergo fragmentation due to surface abrasion, UV degradation, and chemical weathering. These particles are classified based on size as mega plastics, macroplastics, mesoplastics, and microplastics [56]. Microplastics (MPs) and nanoplastics (NPs) are among the most reported emerging toxicants [8]. MPs are defined as artificial

polymers in size ranging between 0.001 and 5 mm, while NPs are artificial polymers with sizes ranging between 1 and 1000 nm [22].

MPs exert a significant impact on soil fauna as they alter soil structure, disrupt nutrient cycling, and influence microbial communities [60]. Understanding and examining the influence of MPs on soil ecosystem health is crucial for sustainable ecosystem management [66]. Terrestrial MPs contamination can indirectly affect marine MPs pollution through surface runoff, groundwater leaching, and organismal transfer [45]. This interconnected pathway of pollution underscores the necessity for comprehensive management strategies that address MPs pollution in both terrestrial and aquatic environments [15].

MPs particles in water bodies impact aquatic invertebrates and fish by disrupting feeding behaviour, fertility, and organ functions [6, 68]. The extent of harm varies based on particle size, dose, and exposure duration. Consequently, MPs present a serious concern for human health, through inhalation, dermal contact, contaminated food and water sources. Inside the body, MPs can trigger oxidative stress, DNA mutilation, organ malfunction, metabolic disorders, immune reactions, neurotoxic effects, and reproductive and developmental toxicity in humans [51]. Epidemiological data also hint at a possible association between MPs exposure and several chronic diseases [11, 37].

Recent studies have demonstrated that MP contamination is not restricted to urban or coastal environments but is increasingly reported from remote, high-altitude, and glacier-fed Himalayan ecosystems. MPs have been detected in Himalayan rivers, lakes, and cryospheric compartments, highlighting the influence of long-range atmospheric transport, increasing tourism, dam-regulated hydrology, and inadequate waste management in mountain regions [3, 33].

Despite the emerging evidence, baseline data on the abundance, composition, and spatial distribution of MPs in glacier-fed river systems of the western Himalaya, particularly across coupled water and sediment matrices, remain extremely limited. This lack of region-specific information constrains a comprehensive understanding of MP sources, transport pathways, and ecological risks in these fragile mountain ecosystems.

The study hypothesised that before estimating the toxic impacts of MPs, it is essential to estimate their presence in the water and sediment of the aquatic system. So, the present study aimed to (i) assess the levels and morphological types and chemical nature of MPs contamination in water and sediment samples from the River Uhl basin through standard methodologies, and (ii) investigate the spatial distribution along different sites of the River Uhl. The findings from the present study will help in implementing effective management strategies, which are imperative for tackling the longstanding ramifications of MPs pollution on aquatic ecosystems [65].

2 Materials and methods

2.1 Study area

The study was conducted in the Kangra district of Himachal Pradesh, focusing on the River Uhl, originating from the Thamsar Glacier in the Dhauladhar range of the Himalayas (31°43′52.4″N, 76°51′15.9″E) (Fig. 1). River Uhl serves as a crucial water source for surrounding communities and is used for hydroelectric power generation. Furthermore,

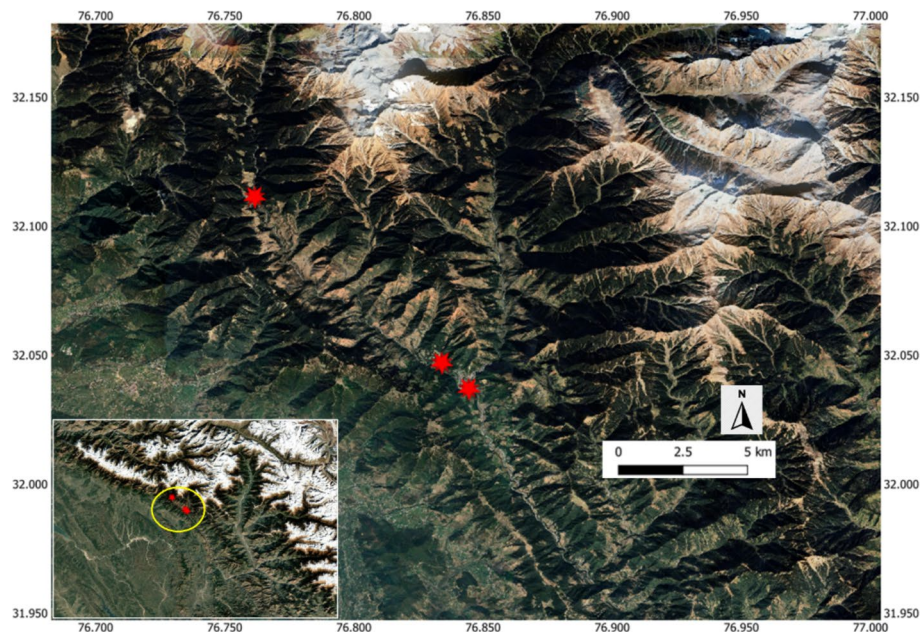


Fig. 1 Map of Study area considered in the Himachal Pradesh

the region attracts tourists seeking solace in its natural splendour, with activities such as trekking, angling, and eco-tourism gaining popularity.

Three distinct sites were selected for water and sediment sampling for the presence of MPs along the River Uhl. The first site (Site I) was Kukar gundha (32.111916° Lat. and 76.760869° Long.) situated approximately 3 kms away from the last village in the area, minimizing human intervention and ensuring a more pristine, contamination-free environment for sampling. The second site (Site II), Zero Point (32.046732° Lat. and 76.833532° Long.), was located 14 kms downstream from the first site, where numerous small tributaries join the main river, and agricultural lands line the riverbanks. This location provides insight into the potential impact of agriculture and domestic activities on water and sediment quality. Finally, the third site (Site III) was Barot (32.037697° Lat. and 76.844332° Long.) positioned at the dam constructed for storing water for a hydro-electric project.

2.2 Sample collection and preparation (water and sediment)

Sampling of water and sediment ($n=3$) was made in November, 2024, from three distinct sites of River Uhl as detailed in Fig. 1.

2.2.1 Water sample preparation

At each sampling site, three independent water samples were collected using pre-cleaned glass bottles (1 L each), following the plastic contamination avoidance protocol recommended by the National Oceanic and Atmospheric Administration (NOAA) [36]. Samples were collected from the side of the river at a depth of 0.5 m below the surface of the water [59]. To ensure sample homogeneity, collected water samples (1 L per jar), total 3 L per site were gently shaken before laboratory processing.

Individual samples were filtered sequentially through stainless-steel sieves of 325 μm and 100 μm mesh size to remove large debris while retaining MP within the defined size range (100 μm –5 mm) [32]. Particles retained were carefully recovered into a clean glass beaker, and volume was adjusted to 100 mL using Milli-Q water to ensure efficient chemical digestion [4]. Organic matter removal was performed using a modified Fenton's reagent method by adding 20 mL of H_2O_2 (30%), followed by gentle heating at 60 °C for 30 min until visible organic material was completely digested. This temperature was selected to minimize potential thermal alteration of MP particles while ensuring effective oxidation.

For the density separation, the sample was added with NaCl (0.5 M) and heated at 75 °C for a minimum of 30 min until complete dissolution of the salt to facilitate salt dissolution without compromising polymer integrity [36]. The suspension was transferred to a density separator and allowed to settle for 24 h. The supernatant was subsequently filtered through a glass microfiber filter (11 μm pore size).

2.2.2 Sediment sample preparation

Three independent sediment samples (1 kg each) were collected from each site from depth of 15 cm using stainless steel shovels, wrapped in aluminium foil, and transported to the laboratory [14, 21]. Collected sediment samples were air-dried at ambient room temperature to remove moisture and establish a dry-weight basis for MPs analysis. After drying, samples were gently homogenized and sieved through stainless-steel meshes (1–5 mm) to remove large debris and macroplastics.

For MP extraction, a 50 g subsample was taken from each 1 kg sediment batch (three subsamples per site) to minimize contamination risk and ensure uniform chemical processing. Each subsample was transferred to a clean glass beaker and mixed with 20 mL of distilled water, followed by mechanical stirring at 2200 rpm for 15 min to disaggregate sediment particles [21].

Density separation was performed by adding 20 mL of saturated NaCl solution (5 M) followed by 20 mL of ZnCl_2 solution (5 M) to increase the solution density and facilitate the flotation of MPs with varying polymer densities. The mixture was allowed to settle overnight (≈ 24 h) to enable clear phase separation.

The supernatant was carefully decanted and filtered using Whatman filter paper (CAT No. 42) into a clean glass beaker. Organic matter digestion was conducted by adding 30% H_2O_2 , followed by incubation at 60 °C for 24 h in an oven to ensure effective oxidation while minimizing thermal damage to MPs [32]. Following digestion, the samples were re-filtered using Whatman CAT No. 42 filter paper for visualization.

2.2.3 MPs visualization

The retained suspected particles on a glass microfiber filter (for water) and a Whatman CAT No. 42 filter paper (for sediment) were visually examined under a MAGNUS CH201 compound microscope equipped with a digital camera at 10 $\times$ and 40 $\times$ magnification. Suspected MPs were manually counted and categorized based on shape and colour. A fibre resembles thin threads and is often coloured, while

fragments are pieces of plastic with diverse shapes. A film is a thin plastic sheet, such as those in grocery bags and food wrappers [48]. Particle size measurements were performed using ImageJ software.

2.3 Quantitation of MPs

To quantitatively assess MPs in water and sediment samples, formula denoted as $MP_{corrected}$ was used [30]:

$$MP_{corrected} = \frac{(MP_{visualized} \times P_{error}) - MP_{blank}}{V \text{ or } M}$$

where, 'MP corrected' represents the final MP concentration, expressed as particles per litre for water samples and particles per kilogram for sediment samples. 'MP visualized' refers to the number of MP particles visually identified under a compound microscope. 'MP blank' represents the average number of particles detected in procedural blanks and was subtracted to correct for possible laboratory contamination. 'V or M' denotes the volume or mass of the analysed sample.

In case of sediment, where 50 g of subsample was used, the values were multiplied with 1000 to represent the units of MP particles per Kg.

2.4 Fourier transform infrared spectroscopy analysis

Fourier Transform Infrared (FT-IR) spectroscopy identifies functional groups of particles based on their unique infrared absorption patterns [35, 49]. Dried glass membrane filter (water samples) and Whatman CAT No. 42 filter paper (sediment samples) containing particles were processed for FT-IR spectra (Agilent Technology Cary 360 h) in a spectrum range of 600–3600 cm^{-1} [2].

2.5 Control of contamination

Steps to reduce the potential plastic contamination included avoiding plastic apparatus and cleansing each apparatus with water and ethanol prior to use. Additionally, samples were meticulously wrapped in aluminium foil to prevent contamination during storage and transportation. To prevent airborne contamination, processing was performed under the laminar flow, procedural blanks were maintained to exclude the contamination, reagents were prefiltered and saved in covered glass containers, and clean-lab practices such as wearing of cotton lab-coat were followed.

2.6 Statistical analysis

For the quantitative assessment of MPs distribution along the three sites, collected samples were analysed statistically. The Kolmogorov–Smirnov and Shapiro–Wilk test indicated that the data was normally distributed. Thus, parametric test- one way ANOVA with post hoc Tukey's test at $P < 0.05$ was applied and analysis was done with SPSS 18.0 statistical software.

3 Results and discussion

3.1 MPs concentration in the sample

The study sites comprised of sites with varying degree of human interferences. The analysis of MPs concentrations along the River Uhl has revealed significant spatial heterogeneity, with Sites II and III exhibiting notably higher levels than Site I. However, the discovery of MPs even in Site I is noteworthy because the site is situated in an area with minimal human intervention and located 2–4 km away from the nearest village, thus providing valuable insight into MPs pervading even remote environments with 60 ± 5 particles L^{-1} in water and 96 ± 7 particles Kg^{-1} in sediment (Table 1). The noteworthy abundance of MPs in Antarctica snow samples with the least human intervention reported an average of 29 particles L^{-1} , with fibres predominating and polyethylene terephthalate (PET) as the primary polymer [5]. At Site II, where anthropogenic activities such as agricultural practices occurred without stringent regulations on plastic usage, MPs contamination was found to be intensified with a count of 100 ± 16 MPs L^{-1} in water and 161 ± 18 MPs Kg^{-1} in sediment (Table 1). Agricultural and household unrecycled plastic can be considered a source of plastic pollution [10, 39]. Meanwhile, Site III, characterized by the presence of dams, demonstrated a different dynamic (Table 1). Here, the water collected and distributed for hydroelectric and drinking purposes contains visible plastic debris, indicating the transport of MPs through hydrological processes. Researchers have found that dams influence the river transport of MP pollution by trapping and redistributing plastic debris downstream, altering flow dynamics, and potentially concentrating MP in reservoirs [18].

Site-wise MPs distribution on the River Uhl, reported Site I, with a comparatively low concentration in both water and sediment samples, with fewer than 100 particles (Table 1). In contrast, Sites II and III fell within the category of moderate to high concentration, with MPs levels ranging from 100 to 500 particles per litre or Kg [5, 27]. The findings of the present study from the River Uhl are largely consistent with previously reported MP investigations in freshwater systems (Table 2). Like the studies conducted by Izlal et al. (2024) in the Surma River and [23, 24] across river water, sediments, and biota, the present study recorded low to moderate MP contamination with clear spatial variability influenced by anthropogenic activities. In all these studies, sediments or sediment matrices exhibited equal or higher MP abundance than surface water, highlighting their role as important sinks for MPs.

Table 1 MPs concentration in three sites

Site Information	Sample type	Concentration (Per Liter or kg)
Site 1	Water	60 ± 5 particles L^{-1}
	Sediment	96 ± 7 particle Kg^{-1}
Site 2	Water	$100^a \pm 16$ particles L^{-1}
	Sediment	$161^a \pm 18$ particle Kg^{-1}
Site 3	Water	$137^{ab} \pm 7$ particles L^{-1}
	Sediment	$195^{ab} \pm 10$ particle Kg^{-1}

MP concentrations in water samples are expressed as particles L^{-1} , while concentrations in sediment samples are expressed as particles Kg^{-1} (dry weight). Values represent site-wise mean concentrations calculated from three samples per site (Mean $\pm$ SD). Superscript alphabets express significant ($p < 0.05$) variations where, 'a'- compared to Site 1, and 'b'- compared to Site 2

Table 2 Comparison of MPs abundance in the present study with worldwide freshwater studies

Study/Location	Sample Type	MP Abundance	Remarks	References
River Uhl, India	Water	60–137 particles L ⁻¹	Moderate contamination; site-wise increase	Present study
River Uhl, India	Sediment	96–195 particles kg ⁻¹	Sediment acts as MP sink	Present study
River Surma, Bangladesh	Water	5–20 particles L ⁻¹	Lower than present study	Rahman et al. [43]
River Surma, Bangladesh	Sediment	360–960 particles kg ⁻¹	Dominant accumulation	Rahman et al. [43]
Urban river & pond, Eastern India	Water	59–100 particles L ⁻¹	Comparable to Site 1–2	Mandal et al. [34]
River Balu & Shitalakshya, Bangladesh	Water	10–50 particles L ⁻¹	Lower water concentration	Roy et al. [47]
River Balu & Shitalakshya, Bangladesh	Sediment	1,200–6,500 particles kg ⁻¹	High urban influence	Roy et al. [47]
Freshwater lake, China	Water	80–120 particles L ⁻¹	Comparable to Site 3	Xiong et al. [64]
Urban rivers, China	Water	0.91 ± 0.20 to 3.40 ± 0.71 particles L ⁻¹	Low concentrations	Zhou et al. [67]
European rivers	Water	~100 particles L ⁻¹	Falls within same range	Eerkes-Medrano et al. [16]
River Ganga, India	Water	8–102 particles L ⁻¹	Lower–moderate range	Varsha et al. [58]
Global freshwater systems	Water	0.001–1000 particles L ⁻¹	Wide global variability	Li et al. [33], Wong et al. [63]

These findings indicate the complex interplay of anthropogenic activities and hydrological processes in shaping MPs contamination patterns. Factors like urbanization, agricultural practices, tourism, and industrial discharge contribute to the MPs pollution in the streams [12].

MPs pollution presents diverse risks to both aquatic ecosystems and human health [40]. The smaller size and large surface area of MPs make them easily ingestible by a wide range of aquatic organisms in aquatic systems. When aquatic organisms ingest MPs, they can experience physical harm, reduced feeding efficiency, and behavioural changes. Further, it can disrupt food webs and ecosystem functions [19, 55]. Notably, MPs act as carriers for persistent organic pollutants (POPs) and heavy metals, which can adhere to their surfaces [7, 13]. These can synergistically increase toxicity, posing enhanced negative impacts on aquatic organisms and human consumers [28].

3.2 MPs characteristics

Under the compound microscope, a preliminary qualitative assessment of MPs identification based on shape and colour was made to evaluate the origin, environmental fate, ecological risk (Fig. 1). MPs that exist in distinct shapes are categorized into fibres, fragments, and films. MPs observed from the study sites were in the size range of 4–11 µm, 15–36 µm, and 5–10 µm for fragments, fibres, and films, respectively (Fig. 2).

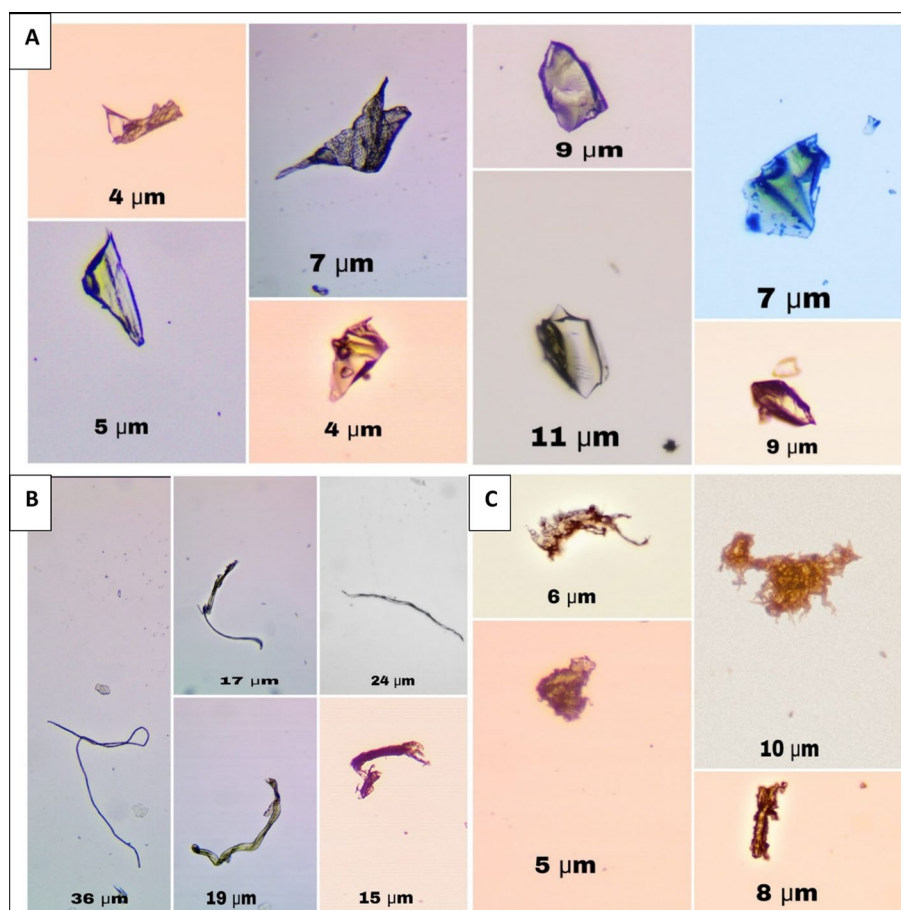


Fig. 2 Representative MPs under compound microscope **A** (Fragment), **B** (Fibre) and **C** (Film) at 40X

3.2.1 Shape

A quantitative assessment of water and sediment samples suggested the pervasive presence of MPs. The water samples collected from the three sites reported a predominance of fragmented MPs, constituting 83.16% followed by fibres (8.42%), and films (8.42%) (Fig. 3A). Similar trend was observed in sediment sample with fragments (87.61%), fibres (6.41%), and films (5.97%) (Fig. 3B). The dominance of fragmented MPs in both water and sediment samples observed in the present study aligns well with findings reported by Izlal et al. [23, 24] and Roy et al. [47], where fragments and fibres were identified as the most prevalent morphologies, indicating secondary fragmentation of larger plastic debris as a major source. In concordance with results, Karthik et al. [26] highlighted the prevalence of MPs fragments (47–50%) compared to fibres (24–27%) and foam (10–19%). These smaller MPs particles may originate from the fragmentation of large pieces, influenced by various physical, biological, and chemical factors [31]. Similar trends of higher fragment abundance have been noted in other geographical locations, such as South China, which has Xijin Wetland Park in Nanning [61] and China Hong Lake and Dongting Lake [62], representative of a broader pattern of MPs pollution. Fibres can be traced to the presence of textile at sites due to waste dumping and sewage flow [39].

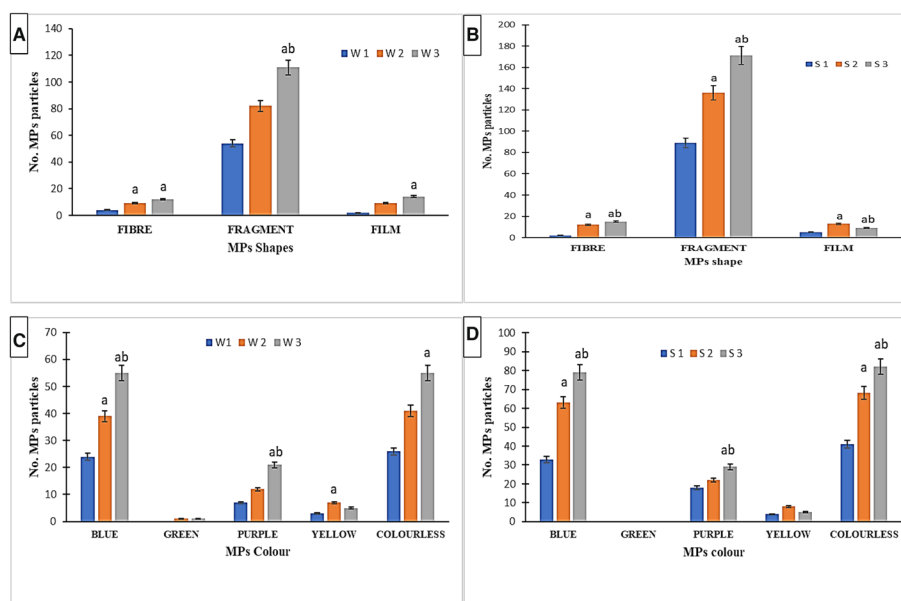


Fig. 3 Quantitative assessment of MPs distribution based on shape (A, B) and colour (C, D) along the different sites of River Uhl. Values were analyzed by one-way ANOVA followed by Tukey Post hoc test. Values are expressed as mean $\pm$ SD. Alphabets express significant ($p < 0.05$) variations where, 'a'—compared to Site 1, and 'b'—compared to Site 2

3.2.2 Colour

Colour-wise assessment of MPs in water samples indicated colourless MPs as the most abundant, accounting for 41.08% of the total, with blue (39.73%), purple (13.47%), and yellow (5.05%) variants (Fig. 3C). The sediment samples collected from the three sites exhibited concordant prevalence of MPs. Colourless MPs dominated in the sediment environment at 42.26%, with blue (38.72%), purple (15.27%), and yellow (3.76%) MPs (Fig. 3D). Interestingly, green coloured particles were observed only in water sample. Accordingly, site III (Barot) followed by site II (Zero point) and lastly site I (Kukar gun-dha) followed the level of MPs contamination. Furthermore, the predominance of colourless and blue MPs in the River Uhl corresponds closely with earlier studies [43, 47], which attributed these colours to common consumer plastics such as packaging materials and bottles. Further, the abundance of colourless MPs suggested possible bleaching of MPs by UV light and other environmental factors [61].

3.3 Fourier transform infrared (FT-IR) spectroscopy

FT-IR analysis was performed independently on MP particles isolated from water and sediment samples to identify their chemical composition. Representative particles were selected based on variations in size, shape, and colour to capture polymer diversity. In water samples, site-specific spectral variations indicated the presence of different polymer types. At Site I, characteristic absorption peaks were recorded at 1638 cm^{-1} , 1716 cm^{-1} , and 3297 cm^{-1} , corresponding to unsaturated polymers, polyesters, and hydroxyl-containing polymers or degradation products (Fig. 3). Observed similar variations in peak positions and intensities at Sites II and III. Sediment samples also exhibited site-dependent FT-IR signatures; for instance, Site I showed prominent peaks at

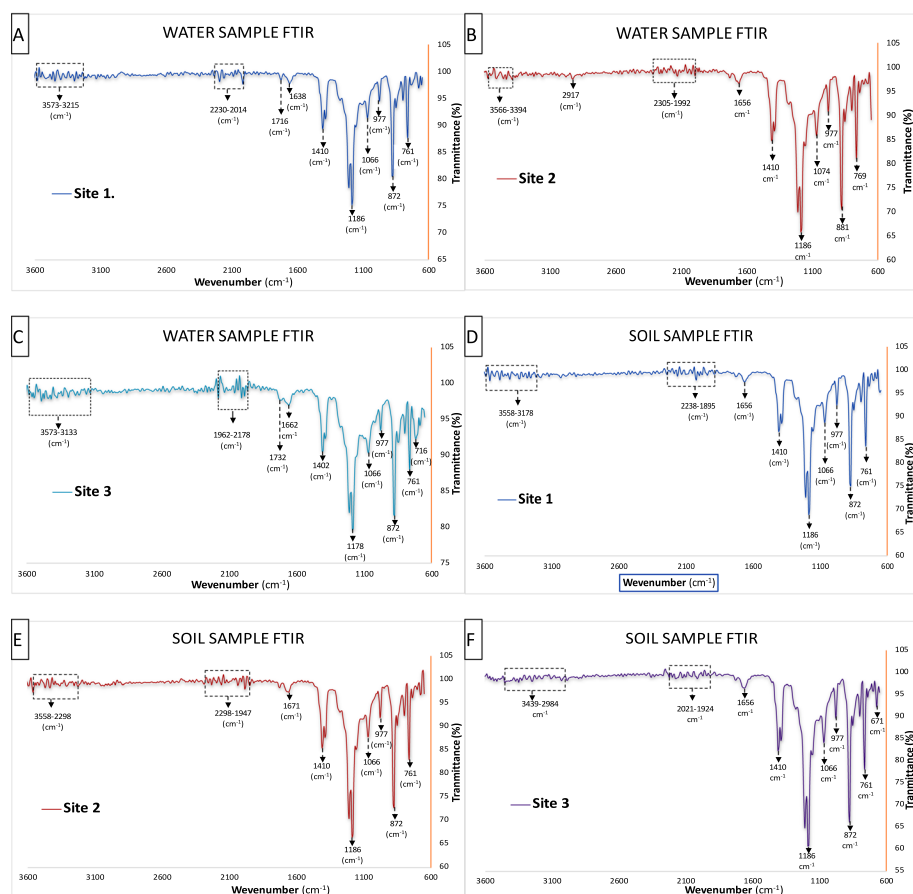


Fig. 4 FT-IR Peak Analysis of water samples (a-c) and sediment samples (d-f) from study sites

1671 cm^{-1} , 2133 cm^{-1} , and 3417 cm^{-1} , suggesting the presence of unsaturated polymers, complex polymer structures, and hydroxyl-containing polymers or degradation products.

Both water and sediment samples shared several absorption peaks at 761 cm^{-1} , 872 cm^{-1} , 977 cm^{-1} , 1066 cm^{-1} , 1186 cm^{-1} , 1410 cm^{-1} , and 3484 cm^{-1} , indicating the occurrence of similar polymeric materials across matrices (Fig. 4). The peak at 761 cm^{-1} represents C–H out-of-plane bending typical of aromatic rings, suggesting the presence of polystyrene or related polymers [20, 53]. Peaks at 872 cm^{-1} and 1410 cm^{-1} correspond to C–H bending vibrations commonly associated with polyethylene and polypropylene [49, 52], while the band at 977 cm^{-1} indicates =C–H bending, which may be attributed to polyvinyl chloride or similar polymers [49], B. [50]. The absorption at 1066 cm^{-1} is characteristic of C–O stretching in polyesters such as polyethylene terephthalate [38, 57], and the peak at 1186 cm^{-1} is related to C–O–C stretching typical of polycarbonates and certain polyesters [20]. Overall, these spectral features confirm the presence of polymers such as polystyrene, polyethylene, polypropylene, polyvinyl chloride, and polyethylene terephthalate reported in freshwater environments [43, 47]. Evidence of weathered and degraded polymer surfaces reported in these studies [23, 24, 47] further supports

the influence of environmental conditions on MP aging, consistent with the spectral variations identified in the present work.

Despite these similarities, distinct site-specific differences in polymer composition and degradation state were evident between water and soil environments, likely influenced by local pollution sources, sediment dynamics, and microbial activity [17]. Comparable spatial variations in sediment samples from Sites II and III are consistent with previous reports documenting MPs in agricultural soil due to plastic mulching, biosolid application, and atmospheric deposition [44, 46].

The detection of polystyrene and polyethylene in soil samples aligns with observations by Nizzetto et al. [39], who highlighted multiple entry pathways of MPs into terrestrial systems, including littering, agricultural practices, and sewage sludge application [10]. The persistence of these polymers in environmental matrices underscores their potential for long-term ecological impacts [18, 39].

4 Conclusion

This study provides baseline evidence of MPs contamination in the glacier-fed River Uhl, Himachal Pradesh. Analysis of water and sediment samples revealed the ubiquitous presence of MPs in both environmental matrices, with fragments as the dominant shape, followed by fibres and films. Predominance of colourless, followed by blue, purple, and yellow MPs particles, reflected commonly used consumer plastics as the primary source. MPs concentrations exhibited marked spatial heterogeneity, with higher abundances at Sites II and III compared to the relatively remote Site I, underscoring the influence of anthropogenic activities and hydrological modifications. Notably, the detection of MPs even at Site I highlights the ability of MPs to disperse and persist in less-disturbed Himalayan environments. FT-IR analysis confirmed the presence of prevalent polymers—polystyrene, polyethylene, and polypropylene, while site-specific spectral variations suggested differences in polymer sources and degradation states between water and sediment. Despite limitations related to spatial coverage, absence of seasonal sampling and small sampling size, this study establishes an important reference point for MP pollution in glacier-fed Himalayan rivers.

Conclusively, addressing MP pollution requires a multifaceted approach encompassing prevention, remediation, and policy interventions. Further, MP source reduction strategies include promotion of sustainable consumption patterns, development of eco-friendly alternatives to plastic, wastewater management, and long-term monitoring strategies.

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Author contributions

A.K. wrote original MS, performed experiments, and analysed results; A.G. analysed and validated results; D.J. worked on methodology and validated results; R.S. conceptualized, supervised and revised the MS.

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Data availability

Data will be made available upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

Authors declare no conflict of interest.

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